QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/46B
21 March 1984
SUPERSEDING
MIL-M-38510/46A
24 May 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER,
DATA SELECTORS/MULTIPLEXERS, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, logic microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Low-power dual, four-input data selector/multiplexer, without enable
02	Low-power eight-input data selector/multiplexer, with enable
03	Low-power quad, two-input data selector/multiplexer, with enable

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to 7.0 V
Input voltage range	-1.5 V at -10 mA to 4.5 V
Storage temperature	-65°C to +150°C
Maximum power dissipation, (P_D) 1/	
Device type 01	150 mW
Device type 02	135 mW
Device type 03	125 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}):	
Case E	0.08°C/mW
Case F	0.09°C/mW
Junction temperature (T_J)	+150°C

1/ Must withstand the added P_D due to short circuit tests (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.7 V
Normalized fanout (each output)	
Low logic level	10 maximum
High logic level	20 maximum
Case operating temperature range (T_C)	-55°C to +125°C.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.2 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.3 Truth tables. The truth tables shall be as specified on figure 2.

3.2.4 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. The "JAN" or "J" certification mark shall not be used.

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Enable input to proper signal (t_{PLH} and t_{PHL})

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.7 V, I_{OH} = -400 µA	All	2.4	---	V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.7 V, I_{OL} = 4.8 mA	All	---	0.3	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -10 mA T_C = 25°C	All	---	-1.5	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	All	0	20	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All	0	100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.3 V	All	-176	-400	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 2/	01	-10	-40	mA
			02,03	-2.5	-25	
Supply current	I_{CC}	V_{CC} = 5.5 V	01		11.5	mA
			02		13.3	
			03		13.2	
Propagation delay time, high-to-low level output from data to 1Y	t_{PHL1}	V_{CC} = 5.0 V C_L = 50 pF R_L = 1.7 kΩ ±5%	01	15	60	ns
Propagation delay time, low-to-high level output from data to 1Y	t_{PLH1}		01	18	80	ns
Propagation delay time, high-to-low level output from data to 2Y	t_{PHL1}		01	10	60	ns
Propagation delay time, low-to-high level output from data to 2Y	t_{PLH1}		01	8	80	ns
Propagation delay time, high-to-low level output from data to $\overline{Y}$	t_{PHL2}		01	8	50	ns
Propagation delay time, low-to-high level output from data to $\overline{Y}$	t_{PLH2}		01	7	45	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time, high-to-low level output from S_0 or S_1 to Y	t_{PHL3}	V_{CC} = 5.0 V R_L = 1.7 kΩ ±5% C_L = 50 pF	01	20	80	ns
Propagation delay time, low-to-high level output from S_0 or S_1 to Y	t_{PLH3}		01	25	90	ns
Propagation delay time, high-to-low level output from S_0 or S_1 to $\overline{Y}$	t_{PHL4}		01	16	80	ns
Propagation delay time, low-to-high level output from S_0 or S_1 to $\overline{Y}$	t_{PLH4}		01	16	70	ns
Propagation delay time, high-to-low level output from S to $\overline{Y}$	t_{PHL1}	V_{CC} = 5.0 V R_L = 1.7 kΩ ±5% C_L = 50 pF	02	15	100	ns
Propagation delay time, low-to-high level output from S to $\overline{Y}$	t_{PLH1}		02	20	120	ns
Propagation delay time, high-to-low level output S to Y	t_{PHL2}		02	25	130	ns
Propagation delay time, low-to-high level output from S to Y	t_{PLH2}		02	25	125	ns
Propagation delay time, high-to-low level output from $\overline{E}$ to $\overline{Y}$	t_{PHL3}		02	14	75	ns
Propagation delay time, low-to-high level output from $\overline{E}$ to $\overline{Y}$	t_{PLH3}		02	15	90	ns
Propagation delay time, high-to-low level output from E to Y	t_{PHL4}		02	15	100	ns
Propagation delay time, low-to-high level output from E to Y	t_{PLH4}		02	20	120	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time, high-to-low level output from data to $\overline{Y}$	t_{PHL5}	V_{CC} = 5.0 V R_L = 1.7 kΩ ±5% C_L = 50 pF	02	10	65	ns
Propagation delay time, low-to-high level output from data to $\overline{Y}$	t_{PLH5}		02	10	70	ns
Propagation delay time, high-to-low level output from data to Y	t_{PHL6}		02	18	100	ns
Propagation delay time, low-to-high level output from data to Y	t_{PLH6}		02	18	90	ns
Propagation delay time, high-to-low level output from S_0 to Y	t_{PHL1}	V_{CC} = 5.0 V R_L = 1.7 kΩ ±5% C_L = 50 pF	03	12	42	ns
Propagation delay time, low-to-high level output from S_0 to Y	t_{PLH1}		03	10	42	ns
Propagation delay time, high-to-low level output from $\overline{E}$ to Y	t_{PHL2}		03	8	35	ns
Propagation delay time, low-to-high level output from $\overline{E}$ to Y	t_{PLH2}		03	8	35	ns
Propagation delay time, high-to-low level output from data to Y	t_{PHL3}		03	5	30	ns
Propagation delay time, low-to-high level output from data to Y	t_{PLH3}		03	8	38	ns

1/ Complete terminal conditions shall be as specified in table III.

2/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)
	Class B devices
Interim electrical test parameters (pre burn-in) (method 5004)	1
Final electrical test parameters (method 5004)	1*, 2, 3, 7, 9
Group A test requirements (method 5005)	1, 2, 3, 7, 9, 10, 11
Group C end-point electrical test parameters (method 5005)	1, 2, 3
Group D end-point electrical test parameters (method 5005)	1, 2, 3
Additional electrical test parameters for group C inspection	None

*PDA applies to subgroup 1 (see 4.2c).

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4 or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameter tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

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4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection.

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical test parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

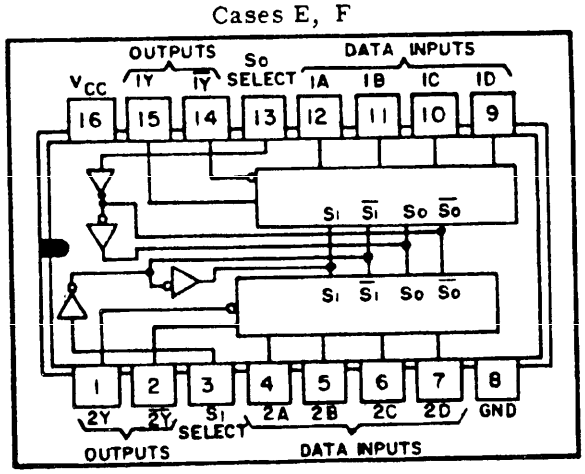
6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

MIL-STD-961D

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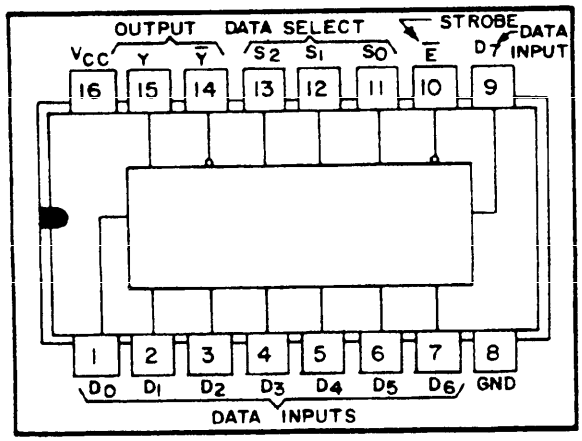
Device type 01

Cases E, F



Device type 02

Cases E, F



Device type 03

Cases E, F

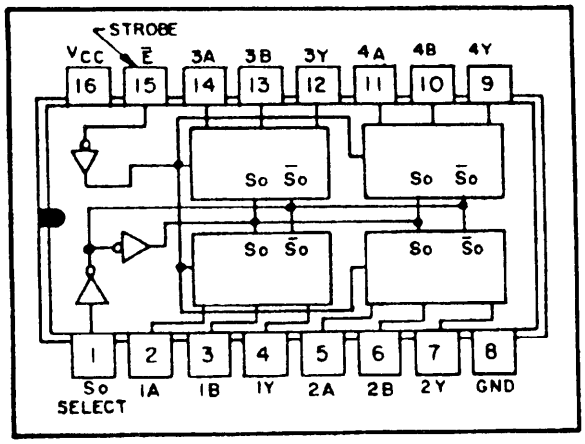


FIGURE 1. Terminal connections.

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Device type 01

Address inputs		Data inputs				Outputs	
S_1	S_0	A	B	C	D	Y	$\overline{Y}$
L	L	L	X	X	X	L	H
L	L	H	X	X	X	H	L
L	H	X	L	X	X	L	H
L	H	X	H	X	X	H	L
H	L	X	X	L	X	L	H
H	L	X	X	H	X	H	L
H	H	X	X	X	L	L	H
H	H	X	X	X	H	L	H

Address inputs S_0 and S_1 are common to both sections.
H = High level, L = Low level, X = Irrelevant.

Device type 02

Inputs												Outputs	
S_2	S_1	S_0	Strobe	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	Y	$\overline{Y}$
X	X	X	H	X	X	X	X	X	X	X	X	L	H
L	L	L	L	L	X	X	X	X	X	X	X	L	H
L	L	L	L	H	X	X	X	X	X	X	X	H	L
L	L	H	L	X	L	X	X	X	X	X	X	L	H
L	L	H	L	X	H	X	X	X	X	X	X	H	L
L	H	L	L	X	X	L	X	X	X	X	X	L	H
L	H	L	L	X	X	H	X	X	X	X	X	H	L
L	H	H	L	X	X	X	L	X	X	X	X	L	H
L	H	H	L	X	X	X	H	X	X	X	X	H	L
H	L	L	L	X	X	X	X	L	X	X	X	L	H
H	L	L	L	X	X	X	X	H	X	X	X	H	L
H	L	H	L	X	X	X	X	X	L	X	X	L	H
H	L	H	L	X	X	X	X	X	H	X	X	H	L
H	H	L	L	X	X	X	X	X	X	L	X	L	H
H	H	L	L	X	X	X	X	X	X	H	X	H	L
H	H	H	L	X	X	X	X	X	X	X	L	L	H
H	H	H	L	X	X	X	X	X	X	X	H	H	L

When used to indicate an input. X = Irrelevant.
H = High level, L = Low level.

Device type 03

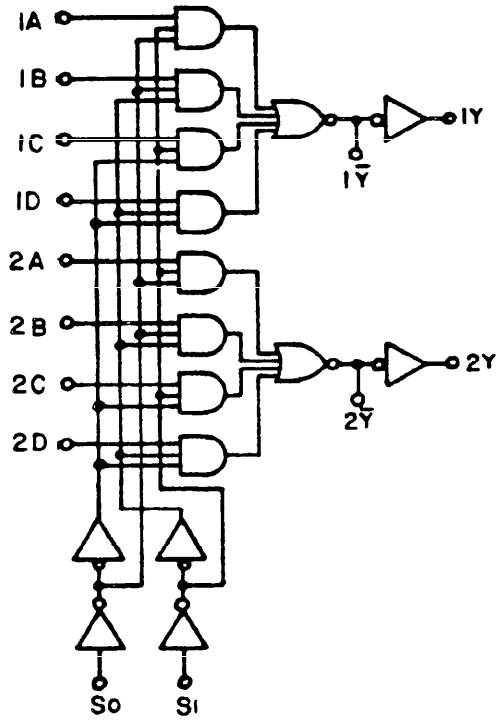
Strobe (enable)	Select input	Data inputs		Output
$\overline{E}$	S_0	A	B	Y
H	X	X	X	L
L	H	X	L	L
L	H	X	H	H
L	L	L	X	L
L	L	H	X	H

Address S_0 and strobe $\overline{E}$ are common to all sections.
H = High level, L = Low level; X = Irrelevant.

FIGURE 2. Truth tables.

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Device type 01



Device type 02

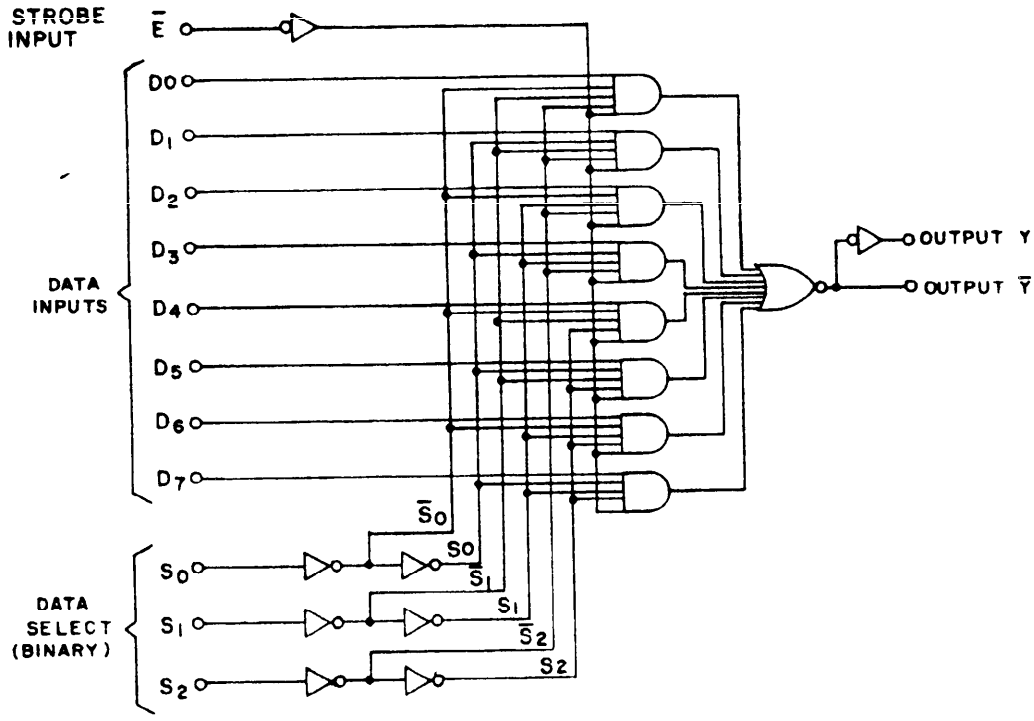


FIGURE 3. Logic diagrams.

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Device type 03

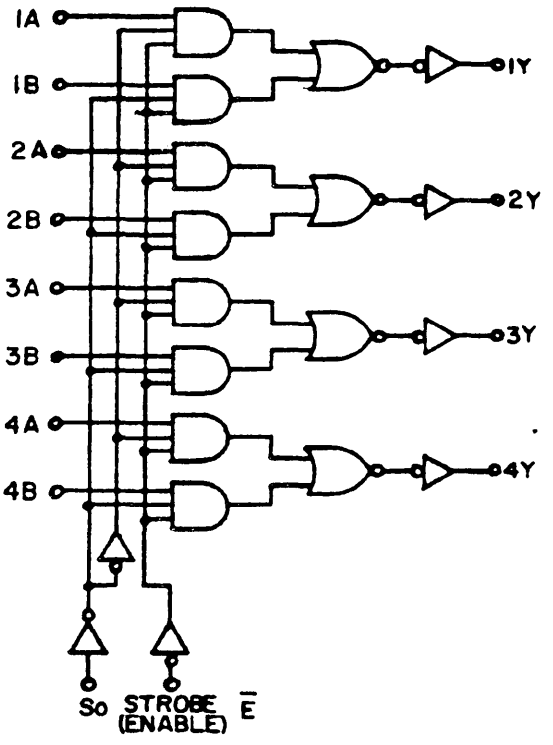
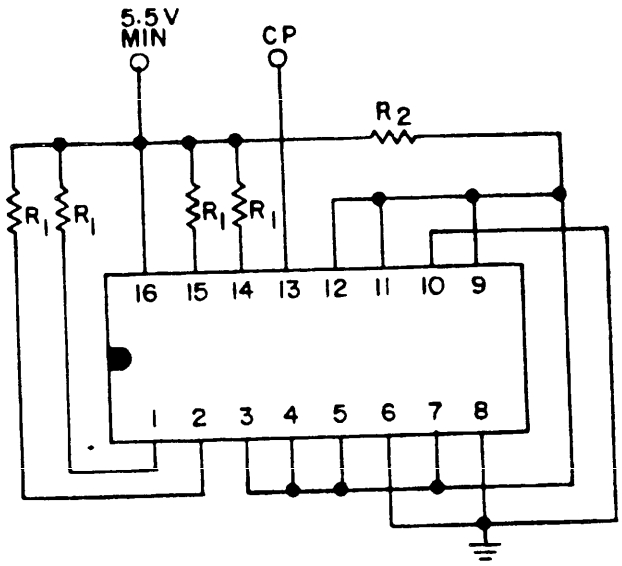


FIGURE 3. Logic diagrams - Continued.

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Device type 01

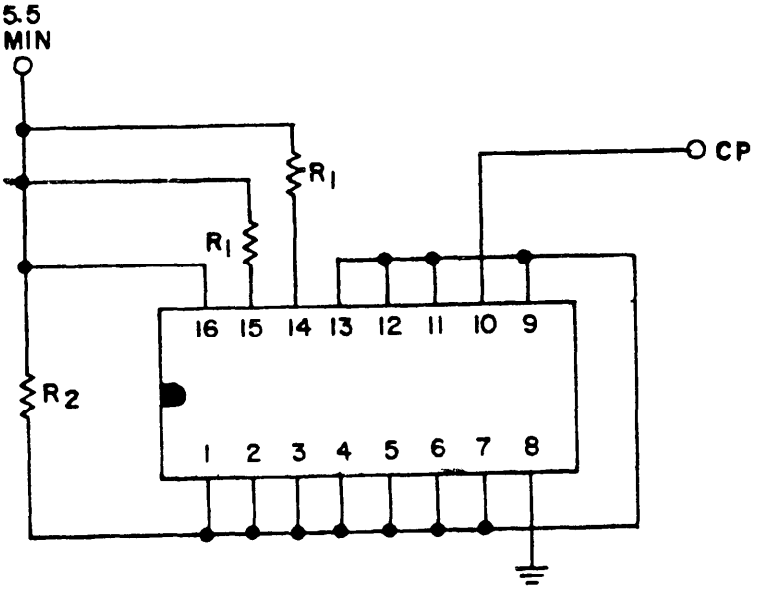


NOTES:
1. CP = 3 V minimum at 100 kHz and 50% duty cycle.
2. R_1 = 325Ω ±5%.
3. R_2 = 16 kΩ ±5%.

FIGURE 4. Burn-in and steady state life test circuits.

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Device type 02

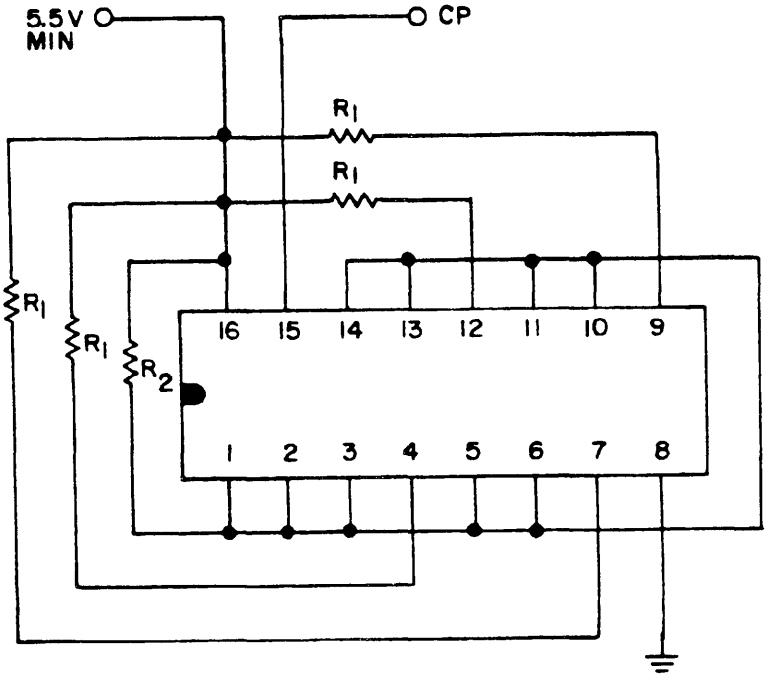


NOTES:
1. CP = 3 V minimum at 100 kHz and 50% duty cycle.
2. R_1 = 325Ω ±5%.
3. R_2 = 16 kΩ ±5%.

FIGURE 4. Burn-in and steady state life test circuits - Continued.

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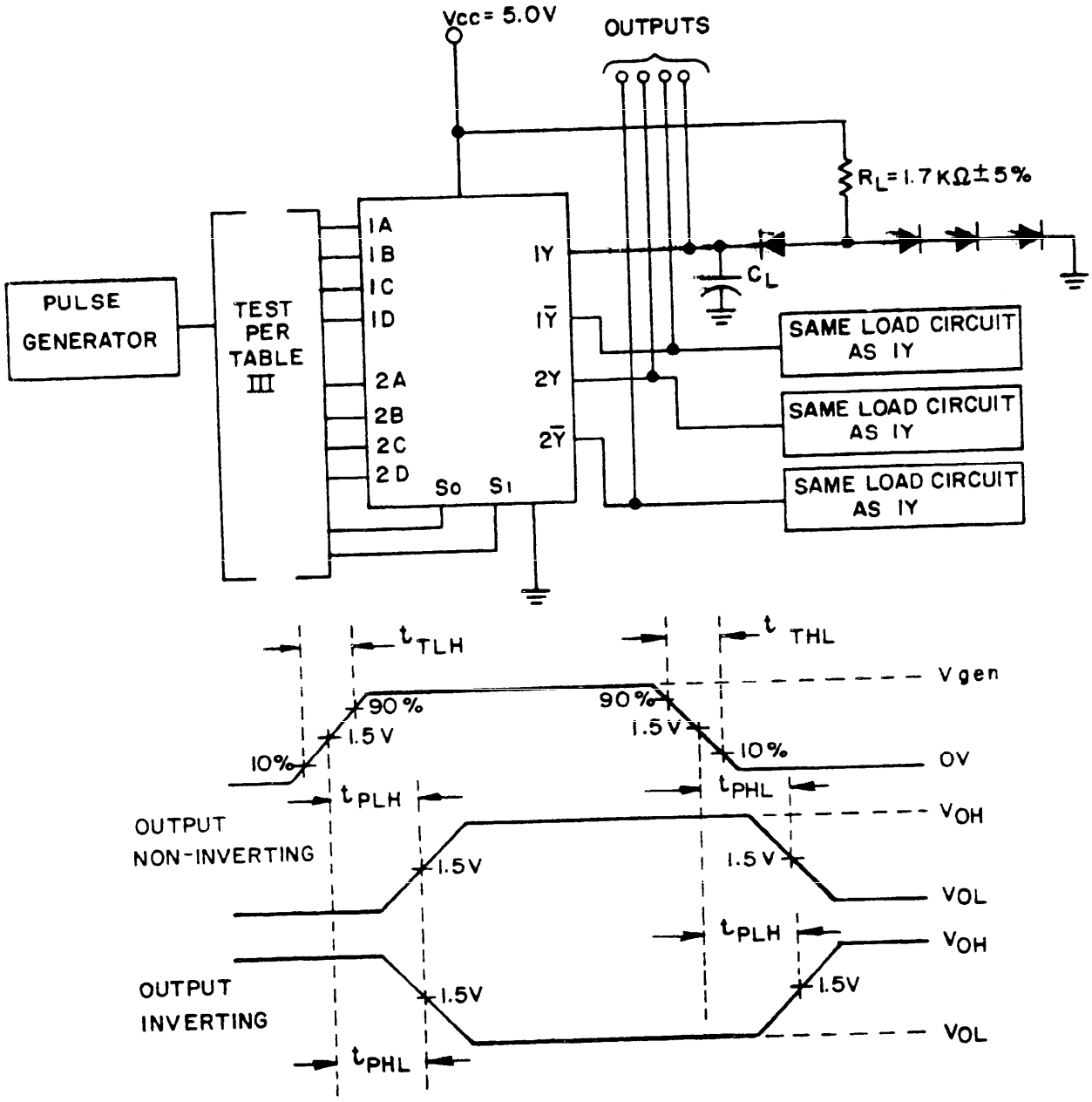
Device type 03



NOTES:
1. CP = 3 V minimum at 100 kHz and 50% duty cycle.
2. R_1 = 325Ω ±5%.
3. R_2 = 16 kΩ ±5%.

FIGURE 4. Burn-in and steady state life test circuits - Continued.

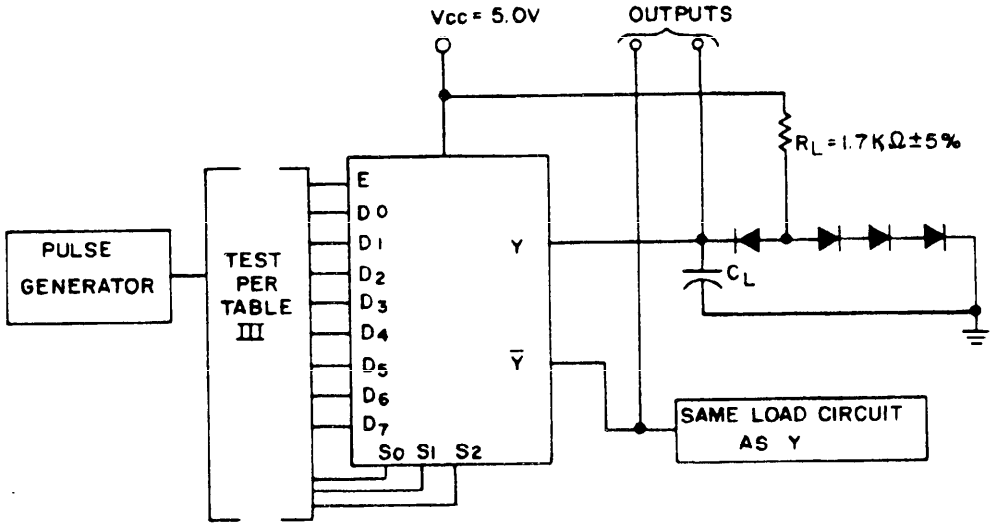
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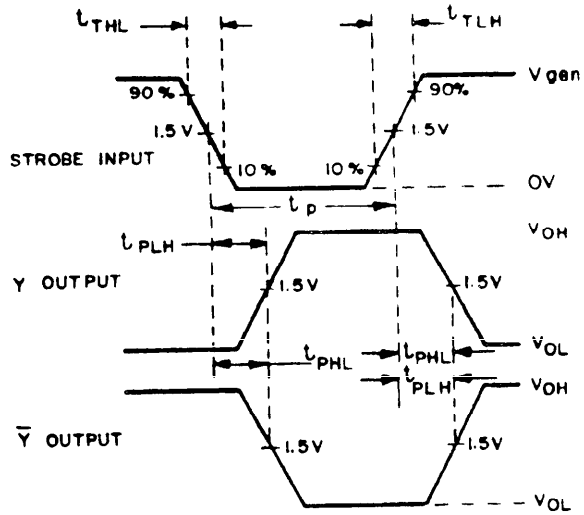


NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3.0 V, PRR = 1 MHz, t_p = 500 ns, $t_{THL} = t_{TLH} \leq 10$ ns, and $Z_{out} \approx 50\Omega$.
2. C_L = 50 pF minimum and includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.

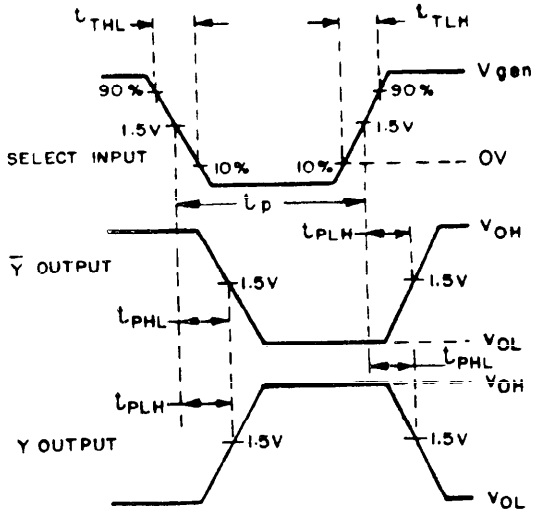
FIGURE 5. Switching time test circuit and waveforms for device type 01.

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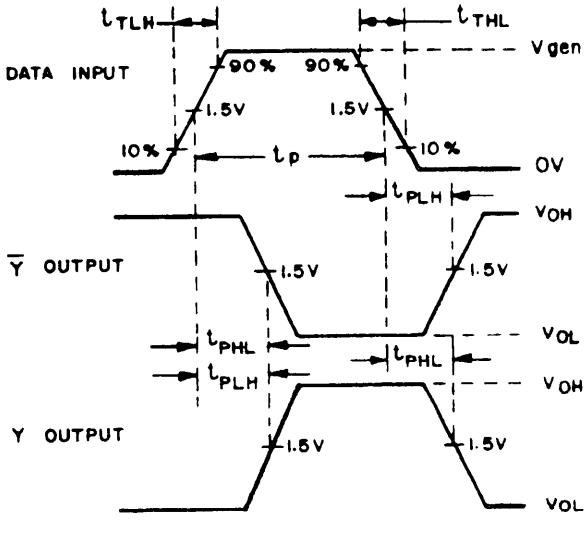




STROBE TO OUTPUT VOLTAGE WAVEFORMS



SELECT INPUT TO OUTPUT VOLTAGE WAVEFORMS

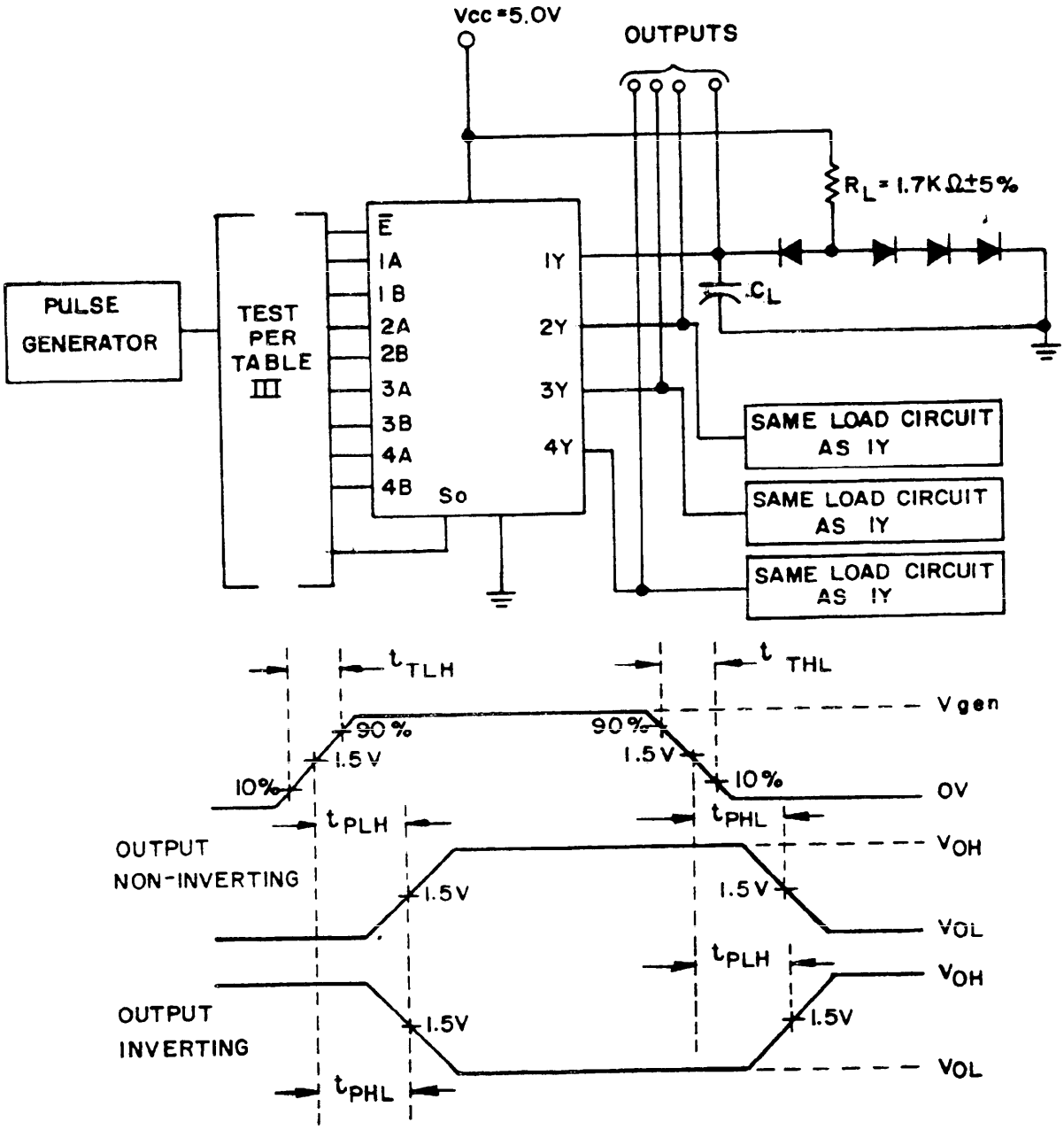


DATA INPUT TO OUTPUT VOLTAGE WAVEFORMS

NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, PRR = 1 MHz, t_p = 500 ns, $t_{THL} = t_{TLH} \leq 10$ ns, and Z_{out} = 50Ω.
2. C_L = 50 pF includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.

FIGURE 5. Switching time test circuit and waveforms for device type 02 - Continued.

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NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, PRR = 1 MHz, t_p = 500 ns, $t_{THL} = t_{TLH} \leq 10$ ns and $Z_{out} \approx 50\Omega$.
2. C_L = 50 pF minimum and includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.

FIGURE 5. Switching time test circuit and waveforms for device type 03 - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	2Y	2$\overline{Y}$	S_1	2A	2B	2C	2D	GND	1D	1C	1B	1A	S_0	1$\overline{Y}$	1Y	V_{CC}		Min	Max	Unit
1 T_C = 25°C	V_{OL}	3007	1			GND					GND				2.0 V	GND	4.8 mA		4.5 V	1$\overline{Y}$		0.3	Vdc
			2			GND										4.5 V				1$\overline{Y}$			
			3			4.5 V							2.0 V	2.0 V		GND				1$\overline{Y}$			
			4			2.0 V						2.0 V				2.0 V				1$\overline{Y}$			
			5			0.7 V									GND	0.7 V		4.8 mA		1Y			
			6		4.8 mA	GND	2.0 V									GND				2$\overline{Y}$			
			7			GND		2.0 V								4.5 V				2$\overline{Y}$			
			8			4.5 V			2.0 V							GND				2$\overline{Y}$			
			9			2.0 V				2.0 V						2.0 V				2$\overline{Y}$			
			10	4.8 mA		0.7 V	GND									0.7 V				2Y		▼	
	V_{OH}	3006	11			GND									0.7 V	GND	-0.4 mA			1$\overline{Y}$	2.4		
			12			GND							0.7 V	0.7 V		4.5 V				1$\overline{Y}$			
			13			4.5 V						0.7 V				GND				1$\overline{Y}$			
			14			4.5 V										4.5 V				1$\overline{Y}$			
			15			GND									4.5 V	GND		-0.4 mA		1Y			
			16		-0.4 mA	GND	0.7 V									GND				2$\overline{Y}$			
			17			GND		0.7 V								4.5 V				2$\overline{Y}$			
			18			4.5 V			0.7 V							GND				2$\overline{Y}$			
			19			4.5 V				0.7 V						4.5 V				2$\overline{Y}$			
			20	-0.4 mA		GND	4.5 V									GND				2Y	▼		
	V_{IC}		21			-10 mA														S_1		-1.5	
			22				-10 mA													2A			
			23					-10 mA												2B			
			24						-10 mA											2C			
			25							-10 mA										2D			
			26									-10 mA								1D			
			27										-10 mA							1C			
			28											-10 mA						1B			
			29												-10 mA					1A			
			30													-10 mA				S_0		▼	▼
	I_{IH1}	3010	31			2.4 V													5.5 V	S_1	0	20	μA
			32			4.5 V	2.4 V									4.5 V				2A			
			33			4.5 V		2.4 V								GND				2B			
			34			GND			2.4 V							4.5 V				2C			
			35							2.4 V						GND				2D			
			36									2.4 V				GND				1D			
			37										2.4 V			4.5 V				1C			
			38			4.5 V								2.4 V		GND				1B			
			39			4.5 V									2.4 V	4.5 V				1A			
			40													2.4 V				S_0		▼	
	I_{IH2}		41			5.5 V														S_1		100	
			42			5.5 V	5.5 V									5.5 V				2A			
			43			5.5 V		5.5 V								GND				2B			
			44			GND			5.5 V							5.5 V				2C			
			45							5.5 V						GND				2D			
			46									5.5 V				GND				1D			
			47										5.5 V			5.5 V				1C			
			48			5.5 V								5.5 V		GND			▼	1B	▼	▼	▼

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01. -Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Max	Unit
				2Y	$2\bar{Y}$	S_1	2A	2B	2C	2D	GND	1D	1C	1B	1A	S_0	$1\bar{Y}$	1Y	V_{CC}				
1 T_C = 25°C	I_{IH2}	3010	49			5.5 V					GND				5.5 V	5.5 V			5.5 V	1A	0	100	µA
	I_{IH2}	3010	50													5.5 V				S_0	0	100	
	I_{IL}	3009	51			0.3 V										GND				S_1	-176	-400	
			52			GND	0.3 V									4.5 V				2A			
			53			GND		0.3 V								GND				2B			
			54			4.5 V			0.3 V							4.5 V				2C			
			55							0.3 V						4.5 V				2D			
			56									0.3 V				GND				1D			
			57										0.3 V			4.5 V				1C			
			58											0.3 V		GND				1B			
			59												0.3 V	4.5 V				1A			
			60			GND										0.3 V				S_0			
	I_{OS}	3011	61			GND						GND	GND	GND	GND	GND	GND	GND		$1\bar{Y}$	-10	-40	mA
			62												4.5 V					1Y			
			63		$\bar{GND}$		GND									GND				$2\bar{Y}$			
			64	GND		GND	4.5 V	GND	GND	GND						GND				2Y			
	I_{CC}	3005	65																	V_{CC}		11.5	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Truth table test		66	L	H	B	B				GND					B			4.5 V	See 1/, 2/			
			67	H	L	B	B									B							
			68	L	H	B	B									A							
			69	H	L	B	B									A							
			70	L	H	A	A	B								B							
			71	H	L	A	A	A								B							
			72	L	H	A	A	A	B							A							
			73	H	L	A	A	A	A	B						B							
			74			B	B	B	B	A						B							
			75			B	B	B	B	A						A							
			76			B	B	B	B	A						B							
			77			B	B	B	B	A						B							
			78			B	B	B	B	A						A							
			79			B	B	B	B	A						B							
			80			B	B	B	B	A						B							
			81			B	B	B	B	A						A							
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PHL1}	3003 Fig. 5	82	OUT		GND	IN				GND	IN	IN	IN	IN	GND		OUT	5.0 V	1A to 1Y	15	45	ns
			83			GND										5.0 V				1B to 1Y			
			84			5.0 V										GND				1C to 1Y			
			85			5.0 V										5.0 V				1D to 1Y			
			86			GND										GND				2A to 2Y	10	35	

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TABLE III. Group A inspection for device type 01. -Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 2Y	2 2$\bar{Y}$	3 S_1	4 2A	5 2B	6 2C	7 2D	8 GND	9 1D	10 1C	11 1B	12 1A	13 S_0	14 1$\bar{Y}$	15 1Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C = 25°C	t_{PHL1} t_{PHL1} t_{PHL1}	3003 Fig. 5	87 88 89	OUT OUT OUT		GND 5.0 V 5.0 V		IN	IN	IN	GND					5.0 V GND 5.0 V			5.0 V	2B to 2Y 2C to 2Y 2D to 2Y	10 10 10	35 35 35	ns
	t_{PLH1}		90 91 92 93 94 95 96 97	OUT		GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	IN	IN	IN	IN		IN	IN	IN	IN	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V		OUT		1A to 1Y 1B to 1Y 1C to 1Y 1D to 1Y 2A to 2Y 2B to 2Y 2C to 2Y 2D to 2Y	20 8 	55 25 	
	t_{PHL2}		98 99 100 101 102 103 104 105		OUT	GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	IN	IN	IN	IN		IN	IN	IN	IN	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	OUT			1A to 1$\bar{Y}$ 1B to 1$\bar{Y}$ 1C to 1$\bar{Y}$ 1D to 1$\bar{Y}$ 2A to 2$\bar{Y}$ 2B to 2$\bar{Y}$ 2C to 2$\bar{Y}$ 2D to 2$\bar{Y}$	10 	35 	
	t_{PLH2}		106 107 108 109 110 111 112 113		OUT	GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	IN	IN	IN	IN		IN	IN	IN	IN	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	OUT			1A to 1$\bar{Y}$ 1B to 1$\bar{Y}$ 1C to 1$\bar{Y}$ 1D to 1$\bar{Y}$ 2A to 2$\bar{Y}$ 2B to 2$\bar{Y}$ 2C to 2$\bar{Y}$ 2D to 2$\bar{Y}$	8 	25 	
	t_{PHL3} t_{PLH3}		114 115 116 117 118 119 120 121	OUT OUT OUT OUT		GND GND IN IN GND GND IN IN	5.0 V 5.0 V 5.0 V 5.0 V	GND GND GND GND	GND GND GND GND			GND GND GND GND	5.0 V 5.0 V 5.0 V 5.0 V	IN IN GND GND IN IN GND GND		OUT OUT OUT OUT		S_0 to 1Y S_0 to 2Y S_1 to 1Y S_1 to 2Y S_0 to 1Y S_0 to 2Y S_1 to 1Y S_1 to 2Y	22 25 	60 65 	
	t_{PHL4} t_{PLH4}		122 123 124 125 126 127 128 129	 OUT OUT OUT OUT		GND GND IN IN GND GND IN IN	5.0 V 5.0 V 5.0 V 5.0 V	GND GND GND GND	 GND GND			 GND GND	GND GND GND GND	5.0 V 5.0 V 5.0 V 5.0 V	IN IN GND GND IN IN GND GND	OUT OUT OUT				S_0 to 1$\bar{Y}$ S_0 to 2$\bar{Y}$ S_1 to 1$\bar{Y}$ S_1 to 2$\bar{Y}$ S_0 to 1Y S_0 to 2$\bar{Y}$ S_1 to 1$\bar{Y}$ S_1 to 1$\bar{Y}$	20 16 	55 50 	
10 T_C = 125°C	t_{PHL1}		130 131 132			GND GND 5.0 V							IN	IN	IN	GND 5.0 V GND		OUT		1A to 1Y 1B to 1Y 1C to 1Y	15 	60 	

See notes at end of device type 01.

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TABLE III. Group A inspection for device type 01.-Continued.

Terminal conditions (pins not designated may be high $\geq$ 2.0 V or low $\leq$ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	2Y	2$\overline{Y}$	S_1	2A	2B	2C	2D	GND	1D	1C	1B	1A	S_0	1$\overline{Y}$	1Y	V_{CC}		Min	Max	Unit
10 T_C = 125°C	t_{PHL1}	3003 Fig 5	133	OUT		5.0 V	IN	IN	IN	IN	GND	IN				5.0 V GND 5.0 V GND 5.0 V		OUT	5.0 V	1D to 1Y 2A to 2Y 2B to 2Y 2C to 2Y 2D to 2Y	15	60	ns
			134			GND																	
			135			GND																	
			136			5.0 V																	
			137			5.0 V																	
	t_{PLH1}		138			GND									IN	GND		$\overline{OUT}$		1A to 1Y 1B to 1Y 1C to 1Y 1D to 1Y 2A to 2Y 2B to 2Y 2C to 2Y 2D to 2Y	20	80	
			139			GND										5.0 V							
			140			5.0 V										GND							
			141			5.0 V										5.0 V							
			142	OUT		GND	IN	IN	IN	IN		IN	IN	IN		GND							
			143			GND										5.0 V							
			144			5.0 V										GND							
			145			5.0 V										5.0 V							
	t_{PHL2}		146			GND									IN	GND	$\overline{OUT}$			1A to 1$\overline{Y}$ 1B to 1$\overline{Y}$ 1C to 1$\overline{Y}$ 1D to 1$\overline{Y}$ 2A to 2$\overline{Y}$ 2B to 2$\overline{Y}$ 2C to 2$\overline{Y}$ 2D to 2$\overline{Y}$	8	40	
			147			GND										5.0 V							
			148			5.0 V										GND							
			149			5.0 V										5.0 V							
			150		OUT	GND	IN	IN	IN	IN		IN	IN	IN		GND							
			151			GND										5.0 V							
			152			5.0 V										GND							
			153			5.0 V										5.0 V							
	t_{PLH2}		154			GND									IN	GND	$\overline{OUT}$			1A to 1$\overline{Y}$ 1B to 1$\overline{Y}$ 1C to 1$\overline{Y}$ 1D to 1$\overline{Y}$ 2A to 2$\overline{Y}$ 2B to 2$\overline{Y}$ 2C to 2$\overline{Y}$ 2D to 2$\overline{Y}$		45	
			155			GND										5.0 V							
			156			5.0 V										GND							
			157			5.0 V										5.0 V							
			158		OUT	GND	IN	IN	IN	IN		IN	IN	IN		GND							
			159			GND										5.0 V							
			160			5.0 V										GND							
			161			5.0 V										5.0 V							
	t_{PHL3}		162			GND								GND	5.0 V	IN		OUT		S0 to 1$\overline{Y}$ S0 to 2$\overline{Y}$ S1 to 1$\overline{Y}$ S1 to 2$\overline{Y}$	20	80	
			163	OUT		GND	5.0 V	GND					GND		5.0 V	IN		OUT					
			164			IN										GND							
			165	OUT		IN	5.0 V		GND							GND							
	t_{PLH3}		166			GND								GND	5.0 V	IN		OUT		S0 to 1$\overline{Y}$ S0 to 2$\overline{Y}$ S1 to 1$\overline{Y}$ S1 to 2$\overline{Y}$	25	90	
			167	OUT		GND	5.0 V	GND					GND		5.0 V	IN		OUT					
			168			IN										GND							
			169	OUT		IN	5.0 V		GND							GND							
	t_{PHL4}		170			GND								GND	5.0 V	IN	OUT			S0 to 1$\overline{Y}$ S0 to 2$\overline{Y}$ S1 to 1$\overline{Y}$ S1 to 2$\overline{Y}$	16	80	
			171		OUT	GND	5.0 V	GND					GND		5.0 V	IN	OUT						
			172			IN										GND							
			173		OUT	IN	5.0 V		GND							GND							
	t_{PLH4}		174			GND								GND	5.0 V	IN	OUT			S0 to 1$\overline{Y}$ S0 to 2$\overline{Y}$ S1 to 1$\overline{Y}$ S1 to 2$\overline{Y}$		70	
			175		OUT	GND	5.0 V	GND					GND		5.0 V	IN	OUT						
			176			IN										GND							
			177		OUT	IN	5.0 V		GND							GND							

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TABLE III. Group A inspection for device type 01. - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 2Y	2 $2\bar{Y}$	3 S_1	4 2A	5 2B	6 2C	7 2D	8 GND	9 1D	10 1C	11 1B	12 1A	13 S_0	14 $1\bar{Y}$	15 1Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
11 T_C = -55°C	t_{PHL1}	3003 Fig. 5	178			GND					GND				IN	GND		OUT	5.0 V	1A to 1Y	15	55	ns
			179			GND								IN		5.0 V				1B to 1Y			
			180			5.0 V							IN			GND				1C to 1Y			
			181			5.0 V						IN				5.0 V				1D to 1Y			
			182	OUT		GND	IN									GND				2A to 2Y			
			183			GND		IN								5.0 V				2B to 2Y			
			184			5.0 V			IN							GND				2C to 2Y			
			185			5.0 V				IN						5.0 V				2D to 2Y			
	t_{PLH1}		186			GND									IN	GND		OUT		1A to 1Y	18	70	
			187			GND								IN		5.0 V				1B to 1Y			
			188			5.0 V							IN			GND				1C to 1Y			
			189			5.0 V						IN				5.0 V				1D to 1Y			
			190	OUT		GND	IN									GND				2A to 2Y			
			191			GND		IN								5.0 V				2B to 2Y			
			192			5.0 V			IN							GND				2C to 2Y			
			193			5.0 V				IN						5.0 V				2D to 2Y			
	t_{PHL2}		194			GND									IN	GND	OUT			1A to 1Y	8	50	
			195			GND								IN		5.0 V				1B to 1Y			
			196			5.0 V							IN			GND				1C to 1Y			
			197			5.0 V						IN				5.0 V				1D to 1Y			
			198		OUT	GND	IN									GND				2A to 2Y			
			199			GND		IN								5.0 V				2B to 2Y			
			200			5.0 V			IN							GND				2C to 2Y			
			201			5.0 V				IN						5.0 V				2D to 2Y			
	t_{PLH2}		202			GND									IN	GND	OUT			1A to 1Y	7	30	
			203			GND								IN		5.0 V				1B to 1Y			
			204			5.0 V							IN			GND				1C to 1Y			
			205			5.0 V						IN				5.0 V				1D to 1Y			
			206		OUT	GND	IN									GND				2A to 2Y			
			207			GND		IN								5.0 V				2B to 2Y			
			208			5.0 V			IN							GND				2C to 2Y			
			209			5.0 V				IN						5.0 V				2D to 2Y			
	t_{PHL3}		210			GND								GND	5.0 V	IN		OUT		S_0 to 1Y	20	80	
			211	OUT		GND	5.0 V	GND								IN				S_0 to 2Y			
			212			IN							GND		5.0 V	GND		OUT		S_1 to 1Y			
			213	OUT		IN	5.0 V		GND							GND				S_1 to 2Y			
	t_{PLH3}		214			GND								GND	5.0 V	IN		OUT		S_0 to 1Y	25	90	
			215	OUT		GND	5.0 V	GND								IN				S_0 to 2Y			
			216			IN							GND		5.0 V	GND		OUT		S_1 to 1Y			
			217	OUT		IN	5.0 V		GND							GND				S_1 to 2Y			
	t_{PHL4}		218			GND								GND	5.0 V	IN	OUT			S_0 to 1Y	18	80	
			219		OUT	GND	5.0 V	GND								IN				S_0 to 2Y			
			220			IN							GND		5.0 V	GND	OUT			S_1 to 1Y			
			221		OUT	IN	5.0 V		GND							GND				S_1 to 2Y			
	t_{PLH4}		222			GND								GND	5.0 V	IN	OUT			S_0 to 1Y	16	70	
			223		OUT	GND	5.0 V	GND								IN				S_0 to 2Y			
			224			IN							GND		5.0 V	GND	OUT			S_1 to 1Y			
			225		OUT	IN	5.0 V		GND							GND				S_1 to 2Y			

1/ A = 4.5 V and B = GND.

2/ Output voltages shall be either:

(a) H = 2.4 volts minimum and L = 0.4 volts maximum when using a high speed checker double comparator, or

(b) H ≥ 1.5 volts and L ≤ 1.5 volts when using a high speed checker single comparator.

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TABLE III Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 D_0	2 D_1	3 D_2	4 D_3	5 D_4	6 D_5	7 D_6	8 GND	9 D_7	10 $\overline{E}$	11 S_0	12 S_1	13 S_2	14 $\overline{Y}$	15 Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	2.0 V	GND	GND	GND	GND	GND	GND	GND	GND	0.7 V	0.7 V	0.7 V	0.7 V	4.8 mA		4.5 V	$\overline{Y}$		0.3	Vdc
			2	GND	2.0 V	GND	↓	↓	↓	↓		↓	↓	2.0 V	2.0 V	↓	↓			↓		↓
			3			2.0 V								0.7 V	0.7 V							
			4			GND	2.0 V							2.0 V	2.0 V							
			5				GND	2.0 V						0.7 V	0.7 V	2.0 V						
			6					GND	2.0 V					2.0 V	0.7 V							
			7						GND	2.0 V				0.7 V	2.0 V							
			8							GND				2.0 V	2.0 V							
			9									2.0 V	GND	GND	GND	GND	2.0 V	4.8 mA		Y			
	V_{OH}	3006	10	4.5 V								GND	2.0 V	GND	GND	GND	-0.4 mA			$\overline{Y}$	2.4		
			11	0.7 V									GND	GND	GND	GND							
			12		0.7 V								4.5 V	4.5 V	GND	↓							
			13			0.7 V							GND	4.5 V	4.5 V	4.5 V							
			14				0.7 V							GND	GND								
			15					0.7 V						4.5 V	4.5 V								
			16						0.7 V					GND	GND								
			17							0.7 V				4.5 V	4.5 V								
			18									0.7 V					0.7 V	-0.4 mA		Y			
			19									4.5 V											
	V_{IC}		20	-10 mA																D_0		-1.5	
			21		-10 mA															D_1			
			22			-10 mA														D_2			
			23				-10 mA													D_3			
			24					-10 mA												D_4			
			25						-10 mA											D_5			
			26							-10 mA										D_6			
			27									-10 mA								D_7			
			28										-10 mA							$\overline{E}$			
			29											-10 mA						S_0			
			30												-10 mA					S_1			
			31													-10 mA				S_2			
	I_{IH1}	3010	32	2.4 V								5.5 V	5.5 V	5.5 V	5.5 V			5.5 V		D_0	0	20	μA
			33		2.4 V								GND	5.5 V	5.5 V					D_1			
			34			2.4 V							5.5 V	GND	GND					D_2			
			35				2.4 V						GND	GND	5.5 V					D_3			
			36					2.4 V					5.5 V	5.5 V	GND					D_4			
			37						2.4 V				GND	GND	GND					D_5			
			38							2.4 V			5.5 V	GND	GND					D_6			
			39									2.4 V	GND	GND	GND					D_7			
			40										2.4 V							$\overline{E}$			
			41											2.4 V						S_0			
			42												2.4 V					S_1			
			43													2.4 V				S_2			
	I_{IH2}		44	5.5 V									5.5 V	5.5 V	5.5 V	5.5 V				D_0		100	
			45		5.5 V									GND	5.5 V					D_1			
			46			5.5 V								5.5 V	GND					D_2			
			47				5.5 V							GND	GND					D_3			

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TABLE III. Group A inspection for device type 02. -Continued.

Terminal conditions (pins not designated may be high > 2.0 V or low < 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 D_0	2 D_1	3 D_2	4 D_3	5 D_4	6 D_5	7 D_6	8 GND	9 D_7	10 $\bar{E}$	11 S_0	12 S_1	13 S_2	14 $\bar{Y}$	15 Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	I_{IH2}	3010	48					5.5 V	5.5 V		GND		5.5 V	5.5 V	5.5 V	GND			5.5 V	D_4	0	100	µA
			49											GND	5.5 V					D_5			
			50							5.5 V				5.5 V	GND					D_6			
			51									5.5 V		GND	GND					D_7			
			52										5.5 V							$\bar{E}$			
			53											5.5 V						S_0			
			54												5.5 V					S_1			
			55																	S_2			
	I_{IL}	3009	56	0.3 V									GND	GND	GND	GND				D_0	-176	-400	
			57		0.3 V								4.5 V	GND	4.5 V					D_1			
			58			0.3 V							GND	4.5 V	GND	4.5 V				D_2			
			59				0.3 V						4.5 V	GND	4.5 V					D_3			
			60					0.3 V					GND	4.5 V	GND					D_4			
			61						0.3 V				4.5 V	GND	4.5 V					D_5			
			62							0.3 V			GND	4.5 V	GND					D_6			
			63										4.5 V	GND	4.5 V					D_7			
			64											4.5 V						$\bar{E}$			
			65												0.3 V					S_0			
			66													0.3 V				S_1			
			67																	S_2			
	I_{OS}	3011	68										4.5 V				GND			$\bar{Y}$	-2.5	-25	mA
	I_{OS}	3011	69	4.5 V									GND	GND	GND	GND		GND		Y	-2.5	-25	mA
	I_{CC}	3005	70	GND	GND	GND	GND	GND	GND	GND		GND		GND	GND	GND				V_{CC}		13.3	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C 25°C	Truth table test		71	B	B	B	B	B	B	B	GND	B	A	B	B	B	B	H	L	4.5 V	See 1/, 2/		
			72	A									B	B	B	B	H	L					
			73		A								B	A	B	B	L	H					
			74			B							B	A	B	B	H	L					
			75			A							B	B	A	B	L	H					
			76				B						B	A	A	B	H	L					
			77				A						B	A	A	B	L	H					
			78					B					B	A	A	A	H	L					
			79					A					B	B	B	A	L	H					
			80						B				B	A	B	A	H	L					
			81						A				B	B	B	A	L	H					
			82							B			B	A	B	A	H	L					
			83							A			B	B	B	A	L	H					
			84								B		B	A	A	A	H	L					
			85								A		B	B	A	A	L	H					
			86									B	B	A	A	A	H	L					
			87									A	B	A	A	A	L	H					
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						

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TABLE III. Group A inspection for device type 02. -Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	D_0	D_1	D_2	D_3	D_4	D_5	D_6	GND	D_7	$\overline{E}$	S_0	S_1	S_2	$\overline{Y}$	Y	V_{CC}		Min	Max	Unit
9 T_C = 25° C	t_{PHL1} t_{PHL1} t_{PHL1} t_{PLH1} t_{PLH1} t_{PLH1}	3003 Fig. 5	88 89 90 91 92 93	GND	5.0 V 5.0 V	5.0 V 5.0 V		5.0 V 5.0 V			GND		GND	IN GND GND IN GND GND	GND IN GND GND IN GND	GND GND IN GND GND IN	OUT		5.0 V	S_0 to $\overline{Y}$ S_1 to $\overline{Y}$ S_2 to $\overline{Y}$ S_0 to $\overline{Y}$ S_1 to $\overline{Y}$ S_2 to $\overline{Y}$	25	90	ns
	t_{PHL2} t_{PHL2} t_{PHL2}		94 95 96		5.0 V	5.0 V	5.0 V							IN GND GND	GND IN GND	GND GND IN		OUT		S_0 to Y S_1 to Y S_2 to Y	30	110 110 110	
	t_{PLH2} t_{PLH2} t_{PLH2}		97 98 99		5.0 V	5.0 V	5.0 V							IN GND	GND IN GND	GND GND IN				S_0 to Y S_1 to Y S_2 to Y		95 95 95	
	t_{PHL3}		100	5.0 V									IN			GND	OUT			$\overline{E}$ to $\overline{Y}$	16	60	
	t_{PLH3}		101	5.0 V													OUT			$\overline{E}$ to $\overline{Y}$	18	75	
	t_{PHL4}		102	5.0 V														OUT		$\overline{E}$ to Y	20	80	
	t_{PLH4}		103	5.0 V														OUT		$\overline{E}$ to Y	25	90	
	t_{PHL5}		104 105 106 107 108 109 110 111	IN	IN	IN	IN	IN	IN	IN		IN	GND	5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	5.0 V 5.0 V GND GND 5.0 V 5.0 V	5.0 V	OUT			D_0 to $\overline{Y}$ D_1 to $\overline{Y}$ D_2 to $\overline{Y}$ D_3 to $\overline{Y}$ D_4 to $\overline{Y}$ D_5 to $\overline{Y}$ D_6 to $\overline{Y}$ D_7 to $\overline{Y}$	10	55	
	t_{PLH5}		112 113 114 115 116 117 118 119	IN	IN	IN	IN	IN	IN	IN		IN		GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	GND 5.0 V 5.0 V 5.0 V 5.0 V				D_0 to $\overline{Y}$ D_1 to $\overline{Y}$ D_2 to $\overline{Y}$ D_3 to $\overline{Y}$ D_4 to $\overline{Y}$ D_5 to $\overline{Y}$ D_6 to $\overline{Y}$ D_7 to $\overline{Y}$	12	60	
	t_{PHL6}		120 121 122 123 124 125 126 127	IN	IN	IN	IN	IN	IN	IN		IN		GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	GND 5.0 V		OUT		D_0 to Y D_1 to Y D_2 to Y D_3 to Y D_4 to Y D_5 to Y D_6 to Y D_7 to Y	20	80	

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TABLE III. Group A inspection for device type 02 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 D_0	2 D_1	3 D_2	4 D_3	5 D_4	6 D_5	7 D_6	8 GND	9 D_7	10 $\overline{E}$	11 S_0	12 S_1	13 S_2	14 $\overline{Y}$	15 Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C = 25° C	t_{PLH6}	3003 Fig. 5	128	IN	IN	IN	IN	IN	IN	IN	GND	IN	GND	GND	GND	GND		OUT	5.0 V	D_0 to Y	20	75	ns
			129											5.0 V	GND					D_1 to Y			
			130											GND	5.0 V					D_2 to Y			
			131											5.0 V	5.0 V					D_3 to Y			
			132											GND	GND	5.0 V				D_4 to Y			
			133											5.0 V	GND					D_5 to Y			
			134											GND	GND					D_6 to Y			
			135											5.0 V	5.0 V					D_7 to Y			
10 T_C = 125° C	t_{PHL1}		136	GND	5.0 V									IN	GND	GND	OUT			S_0 to $\overline{Y}$	15	90	
	t_{PLH1}		137		5.0 V			5.0 V						GND	IN	GND				S_1 to $\overline{Y}$	15	90	
	t_{PHL1}		138											GND	GND	IN				S_2 to $\overline{Y}$	15	90	
	t_{PLH1}		139		5.0 V	5.0 V								IN	GND	GND				S_0 to $\overline{Y}$	20	120	
	t_{PHL1}		140					5.0 V						GND	IN	GND				S_1 to $\overline{Y}$	20		
	t_{PLH1}		141											GND	GND	IN				S_2 to $\overline{Y}$	20		
	t_{PHL2}		142		5.0 V	5.0 V		5.0 V						IN	GND	GND		OUT		S_0 to Y	25		
	t_{PLH2}		143											GND	IN	GND				S_1 to Y			
	t_{PHL2}		144											GND	GND	IN				S_2 to Y		125	
	t_{PLH2}		145		5.0 V	5.0 V		5.0 V					IN	IN	GND	GND				S_0 to Y		125	
	t_{PHL2}		146											GND	IN	GND				S_1 to Y		125	
	t_{PLH2}		147											GND	GND	IN				S_2 to Y			
	t_{PHL3}		148	5.0 V									IN			GND	OUT			$\overline{E}$ to $\overline{Y}$	14	70	
	t_{PLH3}		149														OUT			$\overline{E}$ to $\overline{Y}$	15	85	
	t_{PHL3}		150															OUT		$\overline{E}$ to Y	15	90	
	t_{PLH4}		151															OUT		$\overline{E}$ to Y	20	120	
	t_{PHL5}		152	IN	IN	IN	IN	IN	IN	IN		IN	GND	5.0 V	5.0 V	5.0 V	OUT			D_0 to $\overline{Y}$	10	60	
			153											GND	5.0 V					D_1 to $\overline{Y}$			
			154											5.0 V	GND					D_2 to $\overline{Y}$			
			155											GND	GND					D_3 to $\overline{Y}$			
			156											5.0 V	5.0 V					D_4 to $\overline{Y}$			
			157											GND	GND					D_5 to $\overline{Y}$			
			158											5.0 V	5.0 V					D_6 to $\overline{Y}$			
			159											GND	GND					D_7 to $\overline{Y}$			
	t_{PLH5}		160	IN	IN	IN	IN	IN	IN	IN		IN		5.0 V	5.0 V	5.0 V				D_0 to $\overline{Y}$		70	
			161											GND	5.0 V					D_1 to $\overline{Y}$			
			162											5.0 V	GND					D_2 to $\overline{Y}$			
			163											GND	GND					D_3 to $\overline{Y}$			
			164											5.0 V	5.0 V					D_4 to $\overline{Y}$			
			165											GND	GND					D_5 to $\overline{Y}$			
			166											5.0 V	5.0 V					D_6 to $\overline{Y}$			
			167											GND	GND					D_7 to $\overline{Y}$			

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TABLE III. Group A inspection for device type 02.-Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 D_0	2 D_1	3 D_2	4 D_3	5 D_4	6 D_5	7 D_6	8 GND	9 D_7	10 $\overline{E}$	11 S_0	12 S_1	13 S_2	14 $\overline{Y}$	15 Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10 T_C = 125°C	t_{PHL6}	3003 Fig. 5	168	IN	IN	IN	IN	IN	IN	IN	GND	IN	GND	GND	GND	GND		OUT	5.0 V	D_0 to Y	20	100	ns
			169											5.0 V	GND					D_1 to Y			
			170											GND	5.0 V					D_2 to Y			
			171											5.0 V	5.0 V					D_3 to Y			
			172											GND	GND	5.0 V				D_4 to Y			
			173											5.0 V	GND					D_5 to Y			
			174											GND	5.0 V					D_6 to Y			
			175											5.0 V	5.0 V					D_7 to Y			
	t_{PLH6}		176	IN	IN	IN	IN	IN	IN	IN		IN		GND	GND	GND				D_0 to Y		90	
			177											5.0 V	GND					D_1 to Y			
			178											GND	5.0 V					D_2 to Y			
			179											5.0 V	5.0 V					D_3 to Y			
			180											GND	GND	5.0 V				D_4 to Y			
			181											5.0 V	GND					D_5 to Y			
			182											GND	5.0 V					D_6 to Y			
			183											5.0 V	5.0 V					D_7 to Y			
11 T_C = -55°C	t_{PHL1}		184	GND	5.0 V	5.0 V		5.0 V						IN	GND	GND	OUT			S_0 to $\overline{Y}$	15	100	
	t_{PLH1}		185											GND	IN	GND				S_1 to $\overline{Y}$	15	100	
	t_{PHL1}		186											GND	GND	IN				S_2 to $\overline{Y}$	15	100	
	t_{PLH1}		187		5.0 V	5.0 V		5.0 V						IN	GND	GND				S_0 to $\overline{Y}$	20	120	
	t_{PLH1}		188											GND	IN	GND				S_1 to $\overline{Y}$	20	120	
	t_{PLH1}		189											GND	GND	IN				S_2 to $\overline{Y}$	20	120	
	t_{PHL2}		190		5.0 V	5.0 V		5.0 V						IN	GND	GND		OUT		S_0 to Y	25	130	
	t_{PHL2}		191											GND	IN	GND				S_1 to Y		130	
	t_{PHL2}		192											GND	GND	IN				S_2 to Y		130	
	t_{PLH2}		193		5.0 V	5.0 V		5.0 V						IN	GND	GND				S_0 to Y		125	
	t_{PLH2}		194											GND	IN	GND				S_1 to Y		125	
	t_{PLH2}		195											GND	GND	IN				S_2 to Y		125	
	t_{PHL3}		196	5.0 V									IN			GND	OUT			$\overline{E}$ to $\overline{Y}$	14	75	
	t_{PLH3}		197														OUT	OUT		$\overline{E}$ to Y	15	90	
	t_{PHL4}		198															OUT		$\overline{E}$ to Y	15	100	
	t_{PLH4}		199																	$\overline{E}$ to Y	20	120	
	t_{PHL5}		200	IN	IN	IN	IN	IN	IN	IN		IN	GND	5.0 V	5.0 V	5.0 V	OUT			D_0 to $\overline{Y}$	10	65	
			201											GND	5.0 V					D_1 to $\overline{Y}$			
			202											5.0 V	GND					D_2 to $\overline{Y}$			
			203											GND	GND					D_3 to $\overline{Y}$			
			204											5.0 V	5.0 V					D_4 to $\overline{Y}$			
			205											GND	GND					D_5 to $\overline{Y}$			
			206											GND	5.0 V					D_6 to $\overline{Y}$			
			207											5.0 V	5.0 V					D_7 to $\overline{Y}$			

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TABLE III. Group A inspection for device type 02. -Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Test limits		
			Test No.	D_0	D_1	D_2	D_3	D_4	D_5	D_6	GND	D_7	$\overline{E}$	S_0	S_1	S_2	$\overline{Y}$	Y	V_{CC}	Meas. terminal	Min	Max	Unit
11 T_C = -55° C	t_{PLH5}	3003 Fig. 5	208 209 210 211 212 213 214 215	IN	IN	IN	IN	IN	IN	IN	GND	IN	GND	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	GND 5.0 V	OUT		5.0 V	D_0 to $\overline{Y}$ D_1 to $\overline{Y}$ D_2 to $\overline{Y}$ D_3 to $\overline{Y}$ D_4 to $\overline{Y}$ D_5 to $\overline{Y}$ D_6 to $\overline{Y}$ D_7 to $\overline{Y}$	10	70	ns
	t_{PHL6}		216 217 218 219 220 221 222 223	IN	IN	IN	IN	IN	IN	IN		IN		GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	GND 5.0 V		OUT		D_0 to Y D_1 to Y D_2 to Y D_3 to Y D_4 to Y D_5 to Y D_6 to Y D_7 to Y	18	95	
	t_{PLH6}		224 225 226 227 228 229 230 231	IN	IN	IN	IN	IN	IN	IN		IN		GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	GND GND 5.0 V 5.0 V GND GND 5.0 V 5.0 V	GND 5.0 V				D_0 to Y D_1 to Y D_2 to Y D_3 to Y D_4 to Y D_5 to Y D_6 to Y D_7 to Y		90	

1/ A = 4.5 V and B = GND.

2/ Output voltages shall be either:

(a) H 2.4 volts minimum and L = 0.4 volts maximum when using a high speed checker double comparator, or

(b) H 1.5 volts and L < 1.5 volts when using a high speed checker single comparator.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 S_0	2 1A	3 1B	4 1Y	5 2A	6 2B	7 2Y	8 GND	9 4Y	10 4B	11 4A	12 3Y	13 3B	14 3A	15 $\overline{E}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	2.0 V		0.7 V	4.8 mA		0.7 V	4.8 mA	GND							GND	4.5 V	1Y		0.3	Vdc
			2															0.7 V		2Y			
			3												4.8 mA	0.7 V		GND		3Y			
			4									4.8 mA	0.7 V							4Y			
	V_{OH}	3006	5	GND	2.0 V		-.4 mA													1Y	2.4		
			6	GND		2.0 V	-.4 mA													1Y			
			7	GND				2.0 V		-.4 mA										2Y			
			8	4.5 V					2.0 V	-.4 mA										2Y			
			9	GND											-.4 mA					3Y			
			10	4.5 V											-.4 mA	2.0 V	2.0 V			3Y			
			11	GND								-.4 mA		2.0 V						4Y			
			12	4.5 V								-.4 mA	2.0 V							4Y			
	V_{IC}		13	-10 mA																S_0		-1.5	
			14		-10 mA															1A			
			15			-10 mA														1B			
			16					-10 mA												2A			
			17						-10 mA											2B			
			18														-10 mA			3A			
			19													-10 mA				3B			
			20											-10 mA						4A			
			21										-10 mA							4B			
			22															-10 mA		E			
	I_{IH1}	3010	23															2.4 V	5.5 V	$\overline{E}$	0	20	μA
			24	2.4 V																S_0			
			25		2.4 V															1A			
			26			2.4 V														1B			
			27					2.4 V												2A			
			28						2.4 V											2B			
			29														2.4 V			3A			
			30													2.4 V				3B			
			31											2.4 V						4A			
			32										2.4 V							4B			
	I_{IH2}		33															5.5 V		$\overline{E}$		100	
			34	5.5 V																S_0			
			35		5.5 V															1A			
			36			5.5 V														1B			
			37					5.5 V												2A			
			38						5.5 V											2B			
			39														5.5 V			3A			
			40													5.5 V				3B			
			41											5.5 V						4A			
			42										5.5 V							4B			
	I_{IL}	3009	43	0.3 V														GND		S_0	-176	-400	
			44	GND	0.3 V															1A			
			45	4.5 V		0.3 V														1B			
			46	GND				0.3 V												2A			
			47	4.5 V					0.3 V											2B			

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03.-Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 S_0	2 1A	3 1B	4 1Y	5 2A	6 2B	7 2Y	8 GND	9 4Y	10 4B	11 4A	12 3Y	13 3B	14 3A	15 $\bar{E}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	I_{IL}	3009	48 49 50 51 52	4.5 V GND 4.5 V GND							GND		0.3 V	0.3 V		0.3 V	0.3 V	0.3 V	5.5 V	3A 3B 4A 4B $\bar{E}$	-176	-400	µA
	I_{OS}	3011	53 54 55 56	GND	4.5 V		GND	4.5 V		GND		GND		4.5 V	GND		4.5 V			1Y 2Y 3Y 4Y	-2.5	25	mA
	I_{CC}	3005	57																	V_{CC}		13.2	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C 25°C	Truth table test		58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74	A A B B A A B B A A B B A A B B	B A	B A	L L H L H	B A	B A	L L H L H	GND	L L H L H	B A	B A	L L H L H	B A	B A	A B B B B B B B B B B B B B B B B	4.5 V	See 17, 27			
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PHL1}	3003 Fig. 5	75 76 77 78	IN	GND	5.0 V	OUT	GND	5.0 V	OUT	GND	OUT	5.0 V	GND	OUT	5.0 V	GND	GND	5.0 V	S_0 to 1Y S_0 to 2Y S_0 to 3Y S_0 to 4Y	15	38	ns
	t_{PLH1}		79 80 81 82		GND	5.0 V	OUT	GND	5.0 V	OUT		OUT	5.0 V	GND	OUT	5.0 V	GND			S_0 to 1Y S_0 to 2Y S_0 to 3Y S_0 to 4Y	12	36	
	t_{PHL2}		83	5.0 V		5.0 V	OUT											IN		$\bar{E}$ to 1Y	10	31	

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03. -Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 S_0	2 1A	3 1B	4 1Y	5 2A	6 2B	7 2Y	8 GND	9 4Y	10 4B	11 4A	12 3Y	13 3B	14 3A	15 $\overline{E}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C = 25°C	t_{PHL2} t_{PHL2} t_{PHL2}	3003 Fig. 5	84 85 86	5.0 V					5.0 V	OUT	GND	OUT	5.0 V		OUT	5.0 V		IN	5.0 V	$\overline{E}$ to 2Y $\overline{E}$ to 3Y $\overline{E}$ to 4Y	10	31	ns
	t_{PLH2}		87 88 89 90			5.0 V	OUT		5.0 V	OUT		OUT	5.0 V		OUT	5.0 V				$\overline{E}$ to 1Y $\overline{E}$ to 2Y $\overline{E}$ to 3Y $\overline{E}$ to 4Y			
	t_{PHL3}		91 92 93 94 95 96 97 98	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	IN	IN	OUT OUT	IN	IN	OUT OUT		OUT OUT	IN	IN	OUT OUT	IN	IN	GND		1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	5	26	
	t_{PLH3}		99 100 101 102 103 104 105 106	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	IN	IN	OUT OUT	IN	IN	OUT OUT		OUT OUT	IN	IN	OUT OUT	IN	IN			1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	10	34	
10 T_C = 125°C	t_{PHL1}		107 108 109 110	IN	GND	5.0 V	OUT	GND	5.0 V	OUT		OUT	5.0 V	GND	OUT	5.0 V	GND			S_0 to 1Y S_0 to 2Y S_0 to 3Y S_0 to 4Y	12	42	
	t_{PLH1}		111 112 113 114		GND	5.0 V	OUT	GND	5.0 V	OUT		OUT	5.0 V	GND	OUT	5.0 V	GND			S_0 to 1Y S_0 to 2Y S_0 to 3Y S_0 to 4Y	10	40	
	t_{PHL2}		115 116 117 118	5.0 V		5.0 V	OUT		5.0 V	OUT		OUT	5.0 V		OUT	5.0 V		IN		$\overline{E}$ to 1Y $\overline{E}$ to 2Y $\overline{E}$ to 3Y $\overline{E}$ to 4Y	8	35	
	t_{PLH2}		119 120 121 122		5.0 V	OUT			5.0 V	OUT		OUT	5.0 V		OUT	5.0 V				$\overline{E}$ to 1Y $\overline{E}$ to 2Y $\overline{E}$ to 3Y $\overline{E}$ to 4Y			
	t_{PHL3}		123 124 125 126 127	GND 5.0 V GND 5.0 V GND	IN	IN	OUT OUT	IN	IN	OUT OUT					OUT		IN	GND		1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y	5	30	

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03. -Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	S_0	1A	1B	1Y	2A	2B	2Y	GND	4Y	4B	4A	3Y	3B	3A	$\overline{E}$	V_{CC}		Min	Max	Unit
10 T_C = 125°C	t_{PHL3} t_{PHL3} t_{PHL3}	3003 Fig. 5	128 129 130	5.0 V GND 5.0 V							GND	OUT OUT	IN	IN	OUT	IN		GND	5.0 V	3B to 3Y 4A to 4Y 4B to 4Y	5 5 5	30 30 30	ns
	t_{PLH3}		131 132 133 134 135 136 137 138	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	IN	IN	OUT OUT	IN	IN	OUT OUT		OUT OUT	IN	IN	OUT OUT	IN	IN			1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	8	38	
11 T_C = -55°C	t_{PHL1}		139 140 141 142	IN ↓ ↓ ↓	GND	5.0 V	OUT	GND	5.0 V	OUT		OUT	5.0 V	GND	OUT	5.0 V	GND			S_0 to 1Y S_0 to 2Y S_0 to 3Y S_0 to 4Y	12	42	
	t_{PLH1}		143 144 145 146	↓ ↓ ↓ ↓	GND	5.0 V	OUT	GND	5.0 V	OUT		OUT	5.0 V	GND	OUT	5.0 V	GND		↓	S_0 to 1Y S_0 to 2Y S_0 to 3Y S_0 to 4Y	10	40	
	t_{PHL2}		147 148 149 150	5.0 V ↓ ↓ ↓		5.0 V	OUT		5.0 V	OUT		OUT	5.0 V		OUT	5.0 V		IN		$\overline{E}$ to 1Y $\overline{E}$ to 2Y $\overline{E}$ to 3Y $\overline{E}$ to 4Y	8	35	
	t_{PLH2}		151 152 153 154	↓ ↓ ↓ ↓		5.0 V	OUT		5.0 V	OUT		OUT	5.0 V		OUT	5.0 V		↓		$\overline{E}$ to 1Y $\overline{E}$ to 2Y $\overline{E}$ to 3Y $\overline{E}$ to 4Y	↓	↓	
	t_{PHL3}		155 156 157 158 159 160 161 162	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	IN	IN	OUT OUT	IN	IN	OUT OUT		OUT OUT	IN	IN	OUT OUT	IN	IN	GND		1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	5	30	
	t_{PLH3}		163 164 165 166 167 168 169 170	GND 5.0 V GND 5.0 V GND 5.0 V GND 5.0 V	IN	IN	OUT OUT	IN	IN	OUT OUT		OUT OUT	IN	IN	OUT OUT	IN	IN			1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	8	38	

1/ A = 4.5 V and B = GND.

2/ Output voltages shall be either:

(a) H = 2.4 volts minimum and L = 0.4 volts maximum when using a high speed checker double comparator.

(b) H = 1.5 volts and L = 1.5 volts when using a high speed checker single comparator.

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6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - Electrical ground (common terminal)

V_{IN} - - - - - - - - - - - - Voltage level at an input terminal

I_{IN} - - - - - - - - - - - - Current-flowing into an input terminal

T_C - - - - - - - - - - - - Case temperature

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	93L09
02	93L12
03	93L22, 54L157

6.8 Ordering guidance. Since the qualification requirements have been removed from the specification, orders may be placed immediately.

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0657-5)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/46A	MICROCIRCUITS, DIGITAL, TTL, LOW POWER, DATA SELECTORS/MULTIPLEXERS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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